ModelIt!

Teacher's Guide

Your Sneakers Are Made of Dinosaurs

How Scientists Engineer Materials That Don't Exist in Nature

Quick Reference

Grade Level	Grade 6
NGSS Standard	MS-PS1-3, MS-PS1-6
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: MS-PS1-3, MS-PS1-6

Gather and make sense of information to describe that synthetic materials come from natural resources and impact society. Undertake a design project to construct, test, and modify a device that either releases or absorbs thermal energy by chemical processes.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a computational model showing how polymer chain length, heat treatment, additives, and raw materials determine synthetic material properties.
Disciplinary Core Idea	**PS1.A Structure and Properties of Matter / PS1.B Chemical Reactions** — Each pure substance has characteristic physical and chemical properties that can be used to identify it. Substances react chemically in characteristic ways, and chemical processes can be designed to produce desired materials.
Crosscutting Concept	**Structure and Function** — Students discover that the molecular STRUCTURE of synthetic materials (chain length, arrangement, additives) directly determines their FUNCTION (strength, flexibility, durability).

CAST Testing Alignment

MULTIPLE CHOICE:

Two plastic samples are made from the same polymer. Sample A is rigid and Sample B is flexible. What molecular-level difference most likely explains this?

CONSTRUCTED RESPONSE:

Explain why petroleum (crude oil) can be transformed into thousands of different synthetic materials with different properties. Use the concepts of polymerization, chain length, and additives in your answer.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don’t need to teach all of this — it’s here so you can answer student questions accurately.

Synthetic materials have transformed modern life. From the clothes on your back to the phone in your pocket, most everyday objects contain materials that don't exist in nature — materials engineered by chemists and materials scientists.

From Petroleum to Plastic

Crude oil (petroleum) is a mixture of hydrocarbon molecules formed from ancient marine organisms over millions of years. Refineries separate these molecules by size (distillation). Chemical plants then link small hydrocarbon molecules (monomers) into long chains (polymers) through polymerization. Different monomers and processes create different plastics with vastly different properties.

How Properties Are Controlled

Materials scientists control properties by adjusting: chain length (longer = stronger), branching (more branches = less dense), cross-linking (more = stiffer), crystallinity (more = harder), and additives (plasticizers for flexibility, fillers for strength, pigments for color). The same base polymer — polyethylene — becomes grocery bags (low density, branched) OR bulletproof plates (ultra-high molecular weight, aligned chains).

Environmental Impact

Synthetic materials provide enormous benefits: medical devices, safety equipment, lightweight transportation, and food preservation. However, petroleum-based plastics don't biodegrade easily, creating pollution that persists for hundreds of years. Only 9% of all plastic ever made has been recycled. Scientists are now developing biodegradable polymers from plant-based sources as alternatives.

The Future of Materials

Cutting-edge materials science is producing: self-healing polymers that repair their own damage, shape-memory materials that return to preset shapes, biodegradable plastics from corn and sugarcane, graphene (one atom thick but 200x stronger than steel), and programmable materials that change properties on demand. Understanding molecular structure is the key to all of these innovations.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify polymer chain length, heat treatment, chemical additives, and raw material quality as the key variables determining synthetic material properties.

E **Establish Relationships**
Students determine that longer chains and optimal heat treatment increase strength (positive), while plasticizer additives decrease rigidity (negative relationship with stiffness but positive with flexibility).

V **Visualize & Model**
Students build a model showing how molecular-level decisions during manufacturing determine the final material's macroscopic properties.

E **Evaluate Outcomes**
Students run weak plastic, super strong, and flexible film scenarios to see how the same base chemistry produces radically different materials.

R **Revise & Extend**
Students add cross-linking density or recycled content to explore additional factors in material design.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Look at your shoes. Almost nothing in them exists in nature. So where did it all come from?"*

Watch For

Have students look at their shoes and list the materials they can identify. Most are synthetic.

Slide 2: Learning Objectives

3 min

Say: *"Today we're learning how scientists create materials that nature never imagined."*

Watch For

Have students read objectives. Pre-teach 'synthetic,' 'polymer,' and 'polymerization.'

Slide 3: Big Question

3 min

Say: *"Your shoes, phone case, backpack, and jacket — made from ancient organisms, transformed by chemistry."*

Watch For

Show the journey: oil well → refinery → chemical plant → product. Trace the molecular path.

Slide 4: LEVER Framework

2 min

Say: *"We'll model how scientists dial in exactly the right properties for any material."*

Watch For

Preview LEVER steps. Explain that materials scientists are molecular-level engineers.

Slide 5: Activity 1: Components

8 min

Say: *"What molecular-level variables determine whether a material is strong, flexible, or brittle?"*

Watch For

Guide sorting. Show real material samples (rigid bottle vs. flexible bag vs. stretchy rubber) — all polymers!

Slide 6: Activity 2: Connections

8 min

Say: *"Longer chains = stronger material. More plasticizer = more flexible. You can dial in ANY combination."*

Watch For

Use LEGO chain analogy: short chains pull apart easily, long chains tangle and hold.

Slide 7: Activity 3: Simulation

10 min

Say: *"Let's engineer three materials: weak, super strong, and ultra flexible — from the same starting chemistry!"*

Watch For

Students predict, then run scenarios. Show how dramatically properties change with molecular adjustments.

Slide 8: Discoveries

5 min

Say: *"The same molecules that make a flimsy grocery bag can make a bulletproof vest. It's all about structure."*

Watch For

Discuss environmental impact. Ask: What's the trade-off of materials that last forever?

Slide 9: STEM Challenge

12 min

Say: *"Can you engineer a shoe sole that's strong, flexible, AND sustainable? The planet is counting on you."*

Watch For

Distribute material property targets. Groups design sustainable replacements with scientific justification.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Heat Treatment, Raw Material Quality
- **INTERNAL:** Polymer Chain Length, Chemical Additives

Relationships

- **Polymer Chain Length to Material Strength:** POSITIVE (+) — Longer polymer chains tangle and interlock more effectively, creating stronger intermolecular forces. Ultra-high molecular weight polyethylene has chains so long it can stop bullets.
- **Heat Treatment to Crystallinity:** POSITIVE (+) — Controlled heating allows polymer chains to organize into ordered crystalline regions. More crystallinity means harder, stiffer, more heat-resistant material.
- **Chemical Additives (Plasticizers) to Flexibility:** POSITIVE (+) — Plasticizer molecules insert between polymer chains, pushing them apart and allowing them to slide more easily. This converts rigid PVC pipe into flexible PVC cling wrap — same polymer, different properties.

Key Discoveries

- Longer polymer chains create stronger materials because chains tangle and hold together better
- Heat treatment allows polymer chains to organize into crystalline structures, dramatically increasing strength
- Additives can completely change a material's properties — the same base polymer can become rigid OR flexible
- Almost all synthetic materials trace back to petroleum (ancient organisms), making your sneakers literally made from transformed ancient life

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Plastic comes from chemicals, not nature"	ALL synthetic materials start as natural resources — primarily petroleum (ancient organisms) or sometimes plants. 'Chemical' processes rearrange natural molecules into new structures. Nothing in plastic was created from nothing — it's all rearranged atoms from Earth's resources.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"All plastics are the same"	There are thousands of different plastics with vastly different properties. Polyethylene (PE) is in grocery bags. Polystyrene (PS) is in foam cups. Polycarbonate (PC) is in bulletproof glass. They're all polymers but with different monomers, chain lengths, and structures — like how different LEGO sets use the same basic bricks but create completely different things.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Recycling solves the plastic problem"	Only about 9% of all plastic ever made has been recycled. Many plastics degrade in quality when recycled (downcycling) and can only be recycled 1-2 times. True solutions require reducing plastic use, designing for recyclability from the start, and developing biodegradable alternatives.